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JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A Place to Learn, A Chance to Grow"

Session: 2025-26

MSE-1 EXAMINATION

Course: B. Tech. in Computer Science & Engineering

Semester/ Year: III Semester/ 2nd Year

Subject: Operating System

Subject Code: CS3T001

Max Marks: 20

Date: 04/08/2025

Duration:- 1 Hr.

Instructions to the Students:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

Students should be able:

1. To understand the basics of how operating systems work.
2. To explain how processes and CPU scheduling function in an operating system.
3. To solve common process synchronization problems.
4. To describe memory management concepts, including virtual memory.
5. To Comprehend disk management and the role of file systems in an operating system.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Which of the following is *not* an example of an Operating System? **A)** Linux **B)** Windows **C)** MS Word **D)** macOS	L1/CO1	1
Q.2	Write the following is not a function of an Operating System? **A)** Memory Management **B)** Process Management **C)** Compiler Design **D)** File Management	L3/CO1	1
Q.3	Identify following program is loaded first when the computer starts? **A)** Shell **B)** Bootstrap Program **C)** Application Program **D)** BIOS Driver	L2/CO2	1
Q.4	Which of the following best describes a thread? **A)** A separate application **B)** A lightweight process **C)** A hardware device **D)** A compiler feature	L4/CO2	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Distinguish between Process Vs Thread.(any 4)	L2/CO2	2
Q.6	Define System Call with its types.	L1/CO1	2
Q.7	State the Context Switching with its types.	L1/CO2	2
Q.8	Discuss any two Operating System Structure.	L2/CO1	2

SECTION C

1. **Solve any two questions.**

Q.9 Consider the following table of Arrival Time & Burst Time for three process give below with Time Quantum 2ms. Draw the Gantt Chart & Calculate Competition Time, Turnaround Time & Waiting Time for each process by using Round Robin Scheduling or First Come First Serve Scheduling. L4/CO2 4

Process Id	Arrival Time	Burst Time
P1	0 ms	4 ms
P2	0 ms	2 ms
P3	0 ms	1 ms

Q.10	Describe Operating System with its types.	L6/CO1	4
Q.11	Explain the Process with its different State of Process.	L5/CO2	4